ECA5601 –Advanced Wireless Communication Systems / 5G / 6G
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Ex.No. 14: 5G Throughput vs SNR with Modulation Labels Date:12-8-26

AIM: To simulate the performance of a 5G wireless communication system using MATLAB by analyzing the relationship between Throughput (Mbps), Signal-to-Noise Ratio (SNR) (dB), and different Modulation Schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM).

Objective:

To evaluate the Throughput (Mbps) of a 5G communication system under different Signal-to-Noise Ratio (SNR) conditions.

Procedure:

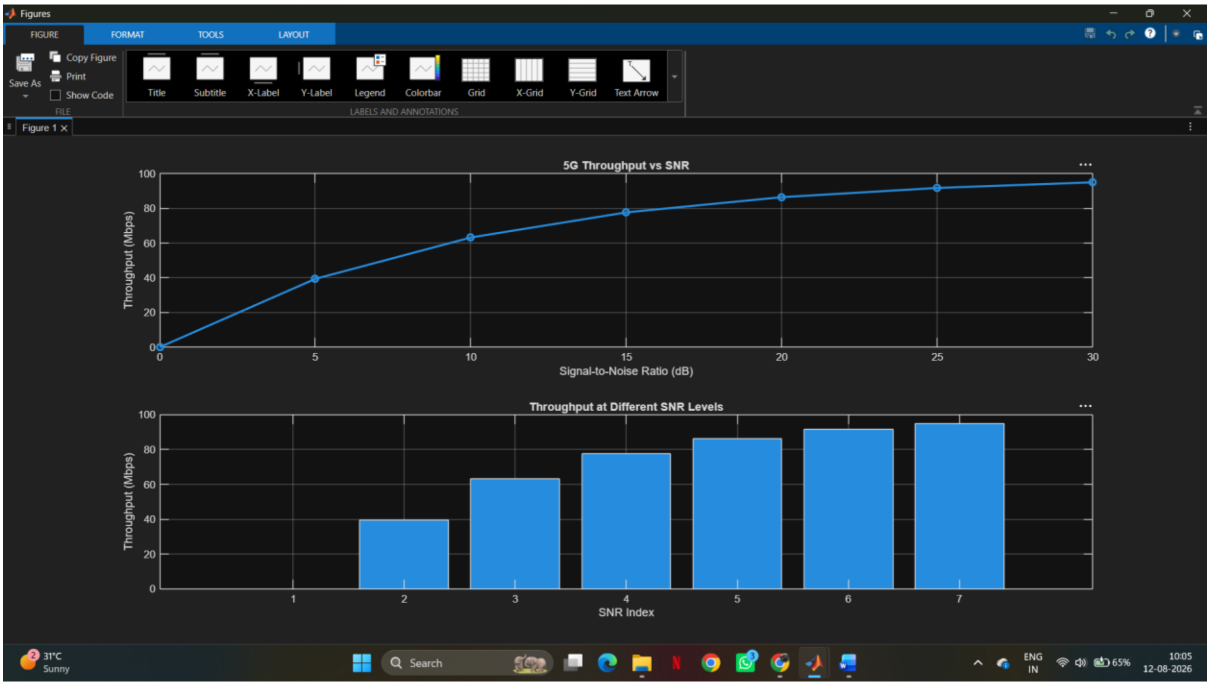
1. Define the simulation parameters such as SNR range, bandwidth, modulation schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM), and transmission settings.
2. Generate random binary data and modulate it using the selected modulation schemes.
3. Simulate the transmission of the modulated signals through an Additive White Gaussian Noise (AWGN) channel with different SNR values.
4. Calculate the Throughput (Mbps) achieved for each modulation scheme at different SNR levels.
5. Analyze the impact of Signal-to-Noise Ratio (SNR) on communication performance and throughput.
6. Compare the performance of QPSK, 16-QAM, 64-QAM, and 256-QAM under identical channel conditions.
7. Plot MATLAB graphs for Throughput vs SNR and SNR vs Modulation Scheme to visualize the system performance.
8. Interpret the results to identify the most suitable modulation scheme for different SNR conditions in a 5G communication system.

Matlab Code :

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation

Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
bar(Throughput)
title('Throughput at Different SNR Levels')
xlabel('SNR Index')
ylabel('Throughput (Mbps)')
grid on
```

Output:



Result:

1.The Throughput (Mbps) of the 5G communication system increased with increasing Signal-to-Noise Ratio (SNR), demonstrating improved data transmission performance.

2.The simulation showed that higher SNR (dB) reduced transmission errors and enhanced the reliability and quality of 5G communication.

3.The comparison of QPSK, 16-QAM, 64-QAM, and 256-QAM revealed that higher-order modulation schemes achieved greater throughput and spectral efficiency but required higher SNR for reliable communication.